

ALTERNATING CURRENT

Important formulas and concepts-

- The electric mains supply in our homes and offices is a voltage that varies like a sine function with time, such a voltage is known as alternating voltage and current driven by it in a circuit is known as alternating current(ac current).

- AC is preferred over DC because

- 1)ac voltages can be efficiently converted from one voltage to another by means of transformers.
- 2)electrical energy can also be transmitted economically over long distances.

❖ AC voltage applied to a resistor-

- Instantaneous Voltage, $V = V_o \sin \omega t$ (V_o is peak value of voltage or Amplitude of oscillating Potential difference)

- Instantaneous Current, $I = (V_o / R) \sin \omega t$
 $I = I_o \sin \omega t$ (I_o is peak value of current)

$$I_o = V_o / R$$

- “Voltage and currents are in phase with each other”.
- Over one complete cycle, the sum of instantaneous current values is zero and average current is also zero.
- Root mean square (rms) current or effective current (I_{rms} or I)

$$I = I_{rms} = I_o / \sqrt{2}$$

Similarly rms voltage,

$$V = V_{rms} = V_o / \sqrt{2}$$

- Average power,

$$\langle P \rangle = VI = V^2 / R$$

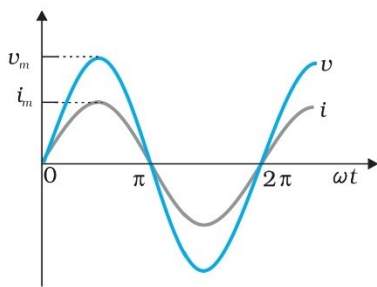


FIGURE 7.2 In a pure resistor, the voltage and current are in phase. The minima, zero and maxima occur at the same respective times.

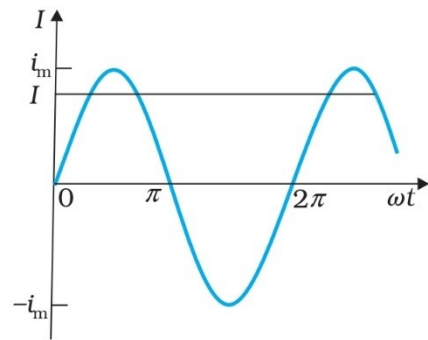


FIGURE 7.3 The rms current I is related to the peak current i_m by $I = i_m / \sqrt{2} = 0.707 i_m$.

❖ AC voltage applied to an inductor-

➤ Inst. Voltage, $V = V_0 \sin \omega t$

➤ Inst. Current, $I = I_0 \sin (\omega t - \pi/2)$

$$I_0 = V_0 / \omega L$$

➤ Inductive reactance,

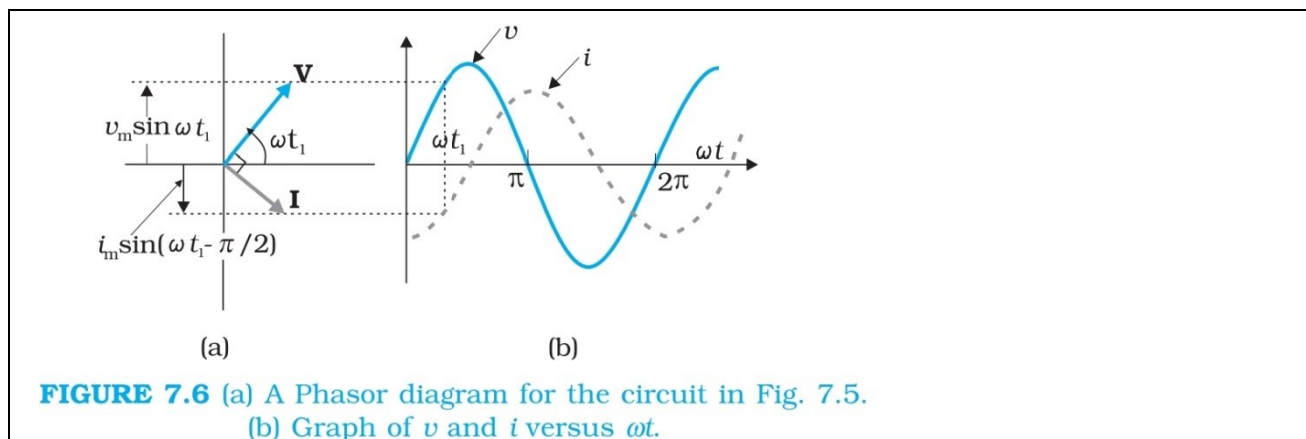
$$X_L = \omega L$$

$$I_0 = V_0 / X_L$$

- It is analogous to resistance.
- Dimensions and unit same as resistance.
- It limits the current in a purely inductive circuit as resistance in purely resistive circuit.
- It is directly proportional to frequency.

➤ “current lags the voltage by $\pi/2$ ”.

➤ Average power over a complete cycle is zero.



❖ AC voltage applied to an capacitor-

➤ Inst. Voltage, $V = V_o \sin \omega t = q/C$

➤ Inst. Current, $I = I_o \sin(\omega t + \pi/2)$

$$I_o = V_o / (1/\omega C)$$

➤ **Capacitive reactance,**

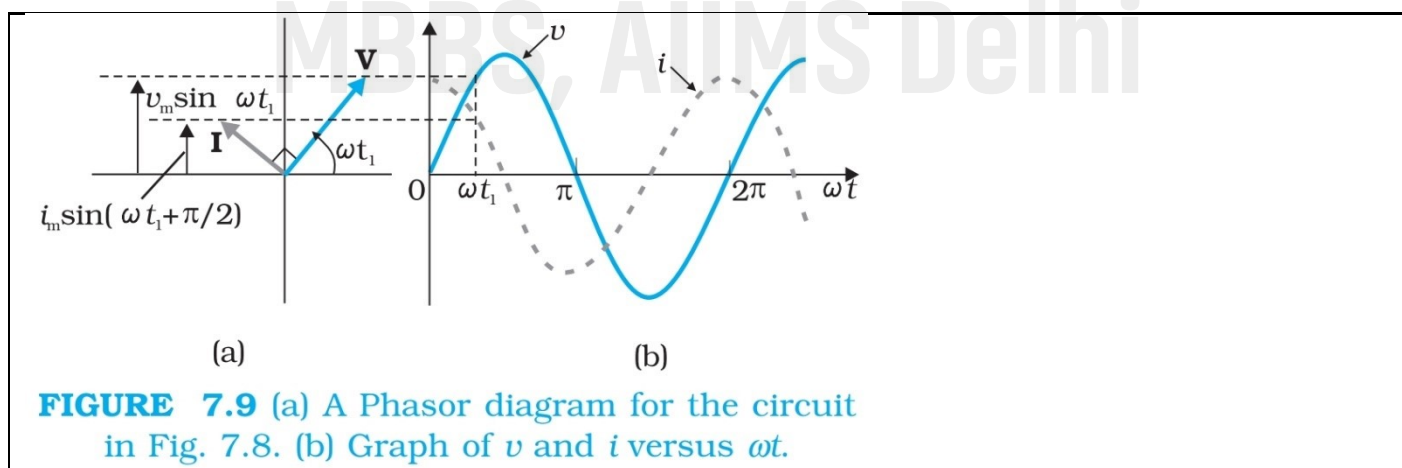
$$X_C = 1/\omega C$$

$$I_o = V_o / X_C$$

- It plays role of resistance.
- Dimensions and unit same as resistance
- It limits the current in a purely capacitive circuit.
- It is inversely proportional to frequency.

➤ **“current is $\pi/2$ ahead of voltage”.**

➤ **The average power over a complete cycle is zero.**



❖ AC voltage applied to a series LCR circuit-

➤ Voltage, $V = V_0 \sin \omega t$

➤ V_R is parallel to I .

V_C and V_L are always along the same line and in opposite directions.

➤ $I_0 = V_0 / \sqrt{R^2 + (X_L - X_C)^2}$

$$I_0 = V_0 / Z$$

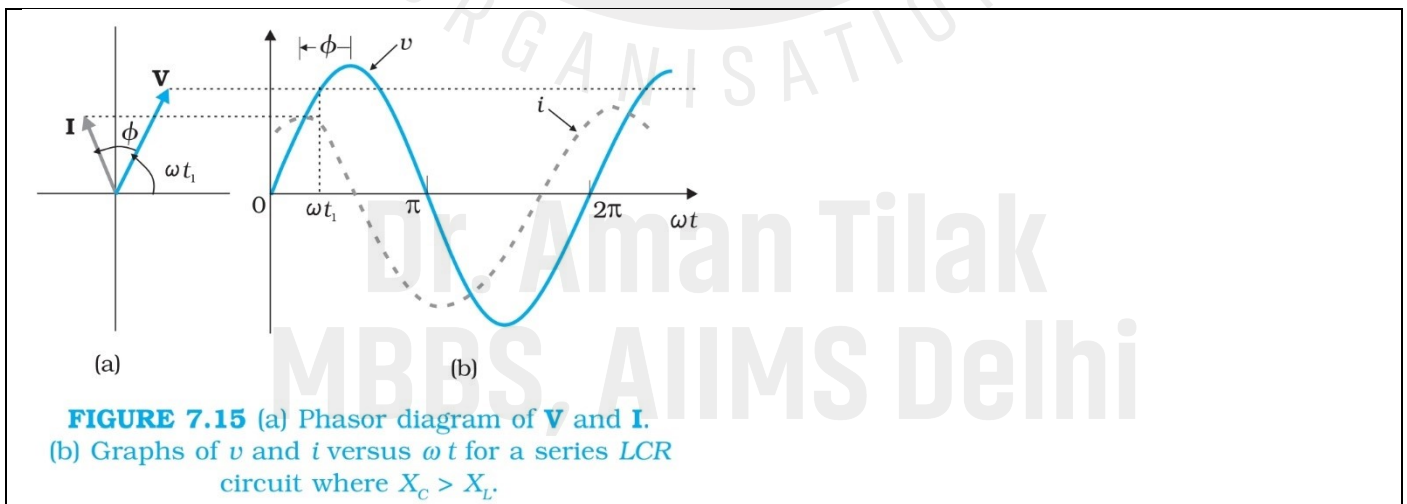
$$\text{Impedance, } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

➤ ϕ is the phase angle b/w V_R or I and V .

$$\tan \phi = V_C - V_L / V_R$$

$$\tan \phi = X_C - X_L / R$$

- if $X_C > X_L$, ϕ is positive and circuit is predominantly capacitive. The current leads the voltage.
- If $X_L > X_C$, ϕ is negative and circuit is predominantly inductive. The voltage leads the current.



❖ Resonance-

➤ At a particular frequency ω_0 , $X_C = X_L$, and the impedance is minimum ($Z = R$), This frequency is known as 'resonant frequency'.

➤ $X_C = X_L$, $1/\omega_0 C = \omega_0 L$

$$\omega_0 = 1/\sqrt{LC}$$

➤ At resonant frequency , current amplitude is maximum,

$$I_0 = V_0 / R$$

➤ Application in ‘the tuning mechanism of radio or a TV set’.

In tuning, the resonant frequency of circuit becomes nearly equal to the frequency of radio signal received.

➤ The resonant frequency is exhibited by a circuit only if both L and C are present in a circuit . we can’t have resonance in a RL or RC circuit.

○ sharpness of resonance-

- if we choose a value of ω for which **current amplitude is $1/\sqrt{2}$ times its maximum value, power dissipated becomes.**

There are two such values of ω , ω_1 and ω_2 , one smaller and one greater than ω_0 .

$$\omega_1 = \omega_0 + \Delta\omega$$

$$\omega_2 = \omega_0 - \Delta\omega$$

- the difference $\omega_1 - \omega_2 = 2\Delta\omega$ is known as ‘ **Bandwidth of circuit**’.
- The quantity $(\omega_0 / 2\Delta\omega)$ is regarded as measure of sharpness of resonance.
- $\Delta\omega = R/2L$

Sharpness of resonance, $\omega_0 / 2\Delta\omega = \omega_0 L/R$

The ratio $\omega_0 L/R$ is also called ‘*quality factor(Q)*’

$$2\Delta\omega = \omega_0 / Q$$

Larger the value of Q, smaller is the value of bandwidth and resonance is sharper.

- $Q = \omega_0 L/R = 1/\omega_0 CR$
- **Less sharp the resonance, less is the selectivity of the circuit.**

If Q is large, i.e. R is low and L is high, the circuit is more selective.

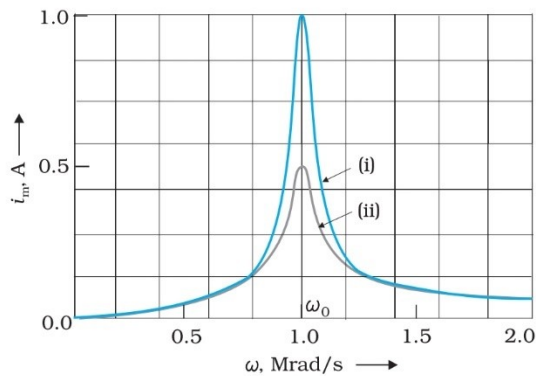


FIGURE 7.16 Variation of i_m with ω for two cases: (i) $R = 100 \, \Omega$, (ii) $R = 200 \, \Omega$, $L = 1.00 \, \text{mH}$.

❖ Power in AC circuit : The Power Factor-

➤ Power factor = $\cos\phi$

1) Resistive circuit: $\phi = 0$, $\cos\phi = 1$
Maximum power dissipation

2) Purely inductive or capacitive circuit: $\phi = \pi/2$, $\cos\phi = 0$
Power dissipated is zero.
Current is known as 'wattless current'

3) LCR circuit,: $\phi = \tan^{-1}(X_C - X_L/R)$
 ϕ may be non zero in RL, RC or LCR circuit
even in such cases, power is dissipated only in resistor.

4) LCR circuit at resonance: $X_C = X_L$, $\phi = 0$, $\cos\phi = 1$
Power dissipated is maximum.

❖ Transformers-

➤ It is a device used to transform an ac voltage from one to another of greater and smaller value.

➤ Based on principle of 'Mutual Induction'.

- Primary and sec. voltages related to no. of turns as,

$$V_S/V_P = N_S/N_P$$

- Currents are related as,

$$I_S/I_P = N_P/N_S$$

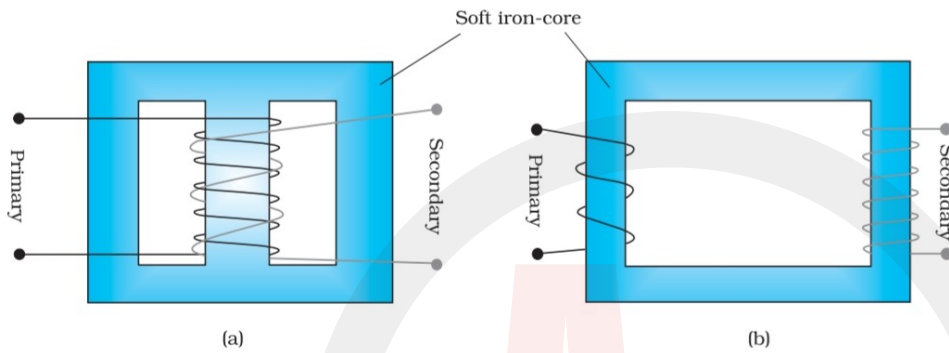


FIGURE 7.20 Two arrangements for winding of primary and secondary coil in a transformer: (a) two coils on top of each other, (b) two coils on separate limbs of the core.

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SUMMARY

1. An alternating voltage $v = v_m \sin \omega t$ applied to a resistor R drives a current $i = i_m \sin \omega t$ in the resistor, $i_m = \frac{v_m}{R}$. The current is in phase with the applied voltage.

2. For an alternating current $i = i_m \sin \omega t$ passing through a resistor R , the average power loss P (averaged over a cycle) due to joule heating is $(1/2) i_m^2 R$. To express it in the same form as the dc power ($P = I^2 R$), a special value of current is used. It is called *root mean square (rms) current* and is denoted by I :

$$I = \frac{i_m}{\sqrt{2}} = 0.707 i_m$$

Similarly, the *rms voltage* is defined by

$$V = \frac{v_m}{\sqrt{2}} = 0.707 v_m$$

We have $P = IV = I^2 R$

3. An ac voltage $v = v_m \sin \omega t$ applied to a pure inductor L , drives a current in the inductor $i = i_m \sin (\omega t - \pi/2)$, where $i_m = v_m / X_L$. $X_L = \omega L$ is called *inductive reactance*. The current in the inductor lags the voltage by $\pi/2$. The average power supplied to an inductor over one complete cycle is zero.
4. An ac voltage $v = v_m \sin \omega t$ applied to a capacitor drives a current in the capacitor: $i = i_m \sin (\omega t + \pi/2)$. Here,

$$i_m = \frac{v_m}{X_C}, X_C = \frac{1}{\omega C} \text{ is called } \textit{capacitive reactance}.$$

The current through the capacitor is $\pi/2$ ahead of the applied voltage. As in the case of inductor, the average power supplied to a capacitor over one complete cycle is zero.

5. For a series RLC circuit driven by voltage $v = v_m \sin \omega t$, the current is given by $i = i_m \sin (\omega t + \phi)$

$$\text{where } i_m = \frac{v_m}{\sqrt{R^2 + (X_C - X_L)^2}}$$

$$\text{and } \phi = \tan^{-1} \frac{X_C - X_L}{R}$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} \text{ is called the } \textit{impedance} \text{ of the circuit.}$$

The average power loss over a complete cycle is given by

$$P = V I \cos \phi$$

The term $\cos \phi$ is called the *power factor*.

6. In a purely inductive or capacitive circuit, $\cos \phi = 0$ and no power is dissipated even though a current is flowing in the circuit. In such cases, current is referred to as a *wattless current*.

7. The phase relationship between current and voltage in an ac circuit can be shown conveniently by representing voltage and current by rotating vectors called *phasors*. A phasor is a vector which rotates about the origin with angular speed ω . The magnitude of a phasor represents the amplitude or peak value of the quantity (voltage or current) represented by the phasor.

The analysis of an ac circuit is facilitated by the use of a phasor diagram.

8. A transformer consists of an iron core on which are bound a primary coil of N_p turns and a secondary coil of N_s turns. If the primary coil is connected to an ac source, the primary and secondary voltages are related by

$$V_s = \left(\frac{N_s}{N_p} \right) V_p$$

and the currents are related by

$$I_s = \left(\frac{N_p}{N_s} \right) I_p$$

If the secondary coil has a greater number of turns than the primary, the voltage is stepped-up ($V_s > V_p$). This type of arrangement is called a *step-up transformer*. If the secondary coil has turns less than the primary, we have a *step-down transformer*.

POINTS TO PONDER

1. When a value is given for ac voltage or current, it is ordinarily the rms value. The voltage across the terminals of an outlet in your room is normally 240 V. This refers to the *rms* value of the voltage. The amplitude of this voltage is

$$v_m = \sqrt{2}V = \sqrt{2}(240) = 340 \text{ V}$$

2. The power rating of an element used in ac circuits refers to its average power rating.
3. The power consumed in an ac circuit is never negative.
4. Both alternating current and direct current are measured in amperes. But how is the ampere defined for an alternating current? It cannot be derived from the mutual attraction of two parallel wires carrying ac currents, as the dc ampere is derived. An ac current changes direction

with the source frequency and the attractive force would average to zero. Thus, the ac ampere must be defined in terms of some property that is independent of the direction of the current. Joule heating is such a property, and there is one ampere of *rms* value of alternating current in a circuit if the current produces the same average heating effect as one ampere of dc current would produce under the same conditions.

5. In an ac circuit, while adding voltages across different elements, one should take care of their phases properly. For example, if V_R and V_C are voltages across R and C , respectively in an RC circuit, then the total voltage across RC combination is $V_{RC} = \sqrt{V_R^2 + V_C^2}$ and not $V_R + V_C$ since V_C is $\pi/2$ out of phase of V_R .
6. Though in a phasor diagram, voltage and current are represented by vectors, these quantities are not really vectors themselves. They are scalar quantities. It so happens that the amplitudes and phases of harmonically varying scalars combine mathematically in the same way as do the projections of rotating vectors of corresponding magnitudes and directions. The 'rotating vectors' that represent harmonically varying scalar quantities are introduced only to provide us with a simple way of adding these quantities using a rule that we already know as the law of vector addition.
7. There are no power losses associated with pure capacitances and pure inductances in an ac circuit. The only element that dissipates energy in an ac circuit is the resistive element.
8. In a RLC circuit, resonance phenomenon occur when $X_L = X_C$ or $\omega_0 = \frac{1}{\sqrt{LC}}$. For resonance to occur, the presence of both L and C elements in the circuit is a must. With only one of these (L or C) elements, there is no possibility of voltage cancellation and hence, no resonance is possible.
9. The power factor in a RLC circuit is a measure of how close the circuit is to expending the maximum power.
10. In generators and motors, the roles of input and output are reversed. In a motor, electric energy is the input and mechanical energy is the output. In a generator, mechanical energy is the input and electric energy is the output. Both devices simply transform energy from one form to another.
11. A transformer (step-up) changes a low-voltage into a high-voltage. This does not violate the law of conservation of energy. The current is reduced by the same proportion.